

Hermit Method

"by using divided difference"

z	$f(z)$	First divided differences	Second divided differences
$z_0 = x_0$	$f[z_0] = f(x_0)$	$f[z_0, z_1] = f'(x_0)$ (given)	
$z_1 = x_0$	$f[z_1] = f(x_0)$	$f[z_1, z_2] = \frac{f[z_2] - f[z_1]}{z_2 - z_1}$ (find from)	$f[z_0, z_1, z_2] = \frac{f[z_1, z_2] - f[z_0, z_1]}{z_2 - z_0}$
$z_2 = x_1$	$f[z_2] = f(x_1)$	$f[z_2, z_3] = f'(x_1)$ (given)	$f[z_1, z_2, z_3] = \frac{f[z_2, z_3] - f[z_1, z_2]}{z_3 - z_1}$
$z_3 = x_1$	$f[z_3] = f(x_1)$	$f[z_3, z_4] = \frac{f[z_4] - f[z_3]}{z_4 - z_3}$	$f[z_2, z_3, z_4] = \frac{f[z_3, z_4] - f[z_2, z_3]}{z_4 - z_2}$
$z_4 = x_2$	$f[z_4] = f(x_2)$	$f[z_4, z_5] = f'(x_2)$ (given)	$f[z_3, z_4, z_5] = \frac{f[z_4, z_5] - f[z_3, z_4]}{z_5 - z_3}$
$z_5 = x_2$	$f[z_5] = f(x_2)$		

$f(z_0, z_1, z_2, z_3)$
 $f(z_1, z_2, z_3, z_4)$
 $f(z_2, z_3, z_4, z_5)$

Simillene

like 3.3

$$H_{2n+1}(x) = f[z_0] + \sum_{k=1}^{2n+1} f[z_0, \dots, z_k](x - z_0)(x - z_1) \dots (x - z_{k-1}).$$

★ Take the Top result ★

Q10 1. Use Theorem 3.9 or Algorithm 3.3 to construct an approximating polynomial for the following data.

a.	x	$f(x)$	$f'(x)$
	8.3	17.56492	3.116256
	8.6	18.50515	3.151762

	z_n	$f(x)$	Deg 1 $f'(x)$	Deg 2 $f(z_0, z_1, z_2), f'(z_1, z_2, z_3)$	Deg 3 $f(z_0, z_1, z_2, z_3)$
	$z_0 = 8.3$	17.56492			
	$z_1 = 8.3$	17.56492	3.116256	$f(z_0, z_1, z_2) = \frac{f(z_1, z_2) - f(z_0, z_1)}{z_0 - z_2}$ $= 0.05948$	
			$\begin{aligned} &= \frac{18.50515 - 17.56492}{8.6 - 8.3} \\ &= 3.1341 \end{aligned}$		
	$z_2 = 8.6$	18.50515	3.151762	$f(z_1, z_2, z_3) = \frac{f(z_2, z_3) - f(z_1, z_2)}{z_3 - z_1}$ $= 0.05887333$	$f(z_0, z_1, z_2, z_3) = \frac{f(z_1, z_2, z_3) - f(z_0, z_1, z_2)}{z_3 - z_0}$ $= -0.00020222$ Could be (-)
	$z_3 = 8.6$	18.50515			

* take the top result

$$M_{2n+1} = f(z_0) + f'(z_0, z_1)(x - z_0) + f(z_0, z_1, z_2)(x - z_0)(x - z_1) + f(z_0, z_1, z_2, z_3)(x - z_0)(x - z_1)(x - z_2)$$

$$M_3 = 17.56492 + 3.116256(x - 8.3) + 0.05948(x - 8.3)(x - 8.3) + (-0.00020222)(x - 8.3)(x - 8.3)(x - 8.6)$$

Q28

2. Use Theorem 3.9 or Algorithm 3.3 to construct an approximating polynomial for the following data.

c.	x	f(x)	f'(x)
	0.1	-0.29004996	-2.8019975
	0.2	-0.56079734	-2.6159201
	0.3	-0.81401972	-2.9734038

$$z_0 = 0.1 \quad -0.29004996$$

$$z_1 = 0.1 \quad -0.29004996$$

$$-2.8019975$$

$$f(z_0, z_1, z_2) = 0.945237$$

$$f(z_1, z_2) = -2.7674738$$

$$f(z_0, z_1, z_2, z_3) = -0.279$$

$$z_2 = 0.2 \quad -0.56079734$$

$$f(z_1, z_2, z_3) = 0.915537$$

$$f(z_0, z_1, z_2, z_3, z_4) = -0.47935$$

$$z_3 = 0.2 \quad -0.56079734$$

$$-2.6159201$$

$$f(z_1, z_2, z_3, z_4) = -0.39287$$

$$f(z_2, z_3, z_4) = 0.836963$$

$$f(z_1, z_2, z_3, z_4, z_5) = -260.4738$$

$$f(z_3, z_4) = -2.5322238$$

$$f(z_2, z_3, z_4, z_5) = -52.48763$$

$$z_4 = 0.3 \quad -0.81401972$$

$$-2.9734038$$

$$f(z_3, z_4, z_5) = -4.4118$$

$$z_5 = 0.3 \quad -0.81401972$$

$$M_5 = f(z_0) + f(z_0, z_1)(x - z_0) + f(z_0, z_1, z_2)(x - z_0)(x - z_1) + f(z_0, z_1, z_2, z_3)(x - z_0)(x - z_1)(x - z_2) + f(z_0, z_1, z_2, z_3, z_4)(x - z_0)(x - z_1)(x - z_2)(x - z_3) + f(z_0, z_1, z_2, z_3, z_4, z_5)(x - z_0)(x - z_1)(x - z_2)(x - z_3)(x - z_4)$$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $-0.29004996 \quad -2.8019975 \quad 0.945237 \quad -0.279 \quad -0.47935 \quad -0.39287 \quad 0.836963 \quad -52.48763 \quad -4.4118 \quad -260.4738$

Q38